

Decay of Heat-Scale Residual Refill at High Frequency

The quantity being measured

The incompressible three-dimensional Navier-Stokes equation is

$$\partial_t u + (u \cdot \nabla)u + \nabla p - \nu \Delta u = 0, \quad \nabla \cdot u = 0,$$

where u is velocity, p is pressure, and $\nu > 0$ is viscosity. A finite-time concentration scenario means that, as time approaches a possible terminal time T_* , the solution's active part is forced into smaller and smaller spatial scales near the terminal region.

Annular or funnel windows are used to watch that terminal region without freezing the observation at one fixed scale. Each window has a terminal end time and a short backward time length. As the windows approach T_* , they look for activity that survives closer and closer to the terminal point.

After a Littlewood-Paley decomposition, smaller spatial scales appear as larger dyadic frequencies. The quantity defined below measures whether residual source can keep refilling those high-frequency shells once heat damping has acted.

Let Δ_j be the Littlewood-Paley projection onto the shell whose frequency size is about 2^j , and write

$$u_j = \Delta_j u.$$

Large j means high frequency and small spatial scale.

Viscosity damps shell j at rate comparable to

$$\nu 2^{2j},$$

where $\nu > 0$ is the viscosity. The natural heat time for shell j is then of order

$$2^{-2j}.$$

A source feeding shell j matters at high frequency through the part that survives this heat-scale damping.

Let $\Phi_j^+(\sigma)$ be the nonnegative residual source feeding shell j at time σ . For two times $t_0 < t < T_*$, define the heat-scale residual refill by

$$D_j^{\text{res}}(t; t_0) = 2^{2j} \int_{t_0}^t e^{-c_0 \nu 2^{2j}(t-\sigma)} \Phi_j^+(\sigma) d\sigma.$$

Here $c_0 > 0$ is a fixed heat-kernel constant. The exponential factor records heat damping from time σ to time t . The outside factor 2^{2j} normalizes the integral to the heat time scale of the shell. Thus $D_j^{\text{res}}(t; t_0)$ measures how much residual source still refills shell j after heat damping.

Dyadic input

The following dyadic source estimate is the input to the decay argument. Let $s > 5/2$, and assume that the velocity field has the uniform Sobolev bound

$$\sup_{t < T_*} \|u(t)\|_{H^s} \leq M,$$

where M is fixed. Assume also that the residual source obeys

$$\Phi_j^+(t) \leq C_s 2^{2j} \|u_j(t)\|_2 \|\Delta_j \mathbb{P} \nabla \cdot (u \otimes u)(t)\|_2.$$

Here C_s depends on s , and $\mathbb{P}$ is the Leray projection onto divergence-free vector fields. All remaining estimates below come from this source bound and the H^s hypothesis.

Proposition (Decay of residual refill under an H^s bound). Under the assumptions above, there is a constant $C'_s > 0$, independent of j , such that

$$D_j^{\text{res}}(t; t_0) \leq C'_s M^3 2^{(3-2s)j}.$$

In particular,

$$D_j^{\text{res}}(t; t_0) \rightarrow 0 \quad \text{as } j \rightarrow \infty.$$

Proof

The proof is a power count. Each step records the power of 2^j that remains at shell j .

First, the H^s bound controls one shell. On shell j , applying s derivatives costs 2^{sj} . Hence

$$2^{sj} \|u_j(t)\|_2 \leq M.$$

Dividing by 2^{sj} gives

$$\|u_j(t)\|_2 \leq M 2^{-sj}.$$

Second, the nonlinear term has one derivative. Since $s > 5/2$, the space $H^s(\mathbb{R}^3)$ is an algebra. Thus

$$\|u \otimes u\|_{H^s} \leq C_s \|u\|_{H^s}^2 \leq C_s M^2.$$

Taking the j -th shell gives

$$\|\Delta_j(u \otimes u)\|_2 \leq C_s M^2 2^{-sj}.$$

The operator $\nabla \cdot$ contributes one derivative, and one derivative on shell j costs 2^j . The Leray projection $\mathbb{P}$ has order zero. Thus

$$\|\Delta_j \mathbb{P} \nabla \cdot (u \otimes u)(t)\|_2 \leq C_s M^2 2^j 2^{-sj}.$$

Equivalently,

$$\|\Delta_j \mathbb{P} \nabla \cdot (u \otimes u)(t)\|_2 \leq C_s M^2 2^{(1-s)j}.$$

Insert the two shell bounds into the source estimate:

$$\Phi_j^+(t) \leq C_s 2^{2j} (M 2^{-sj}) (M^2 2^{(1-s)j}).$$

The constants multiply as

$$M \cdot M^2 = M^3.$$

The dyadic powers multiply as

$$2^{2j} 2^{-sj} 2^{(1-s)j} = 2^{(3-2s)j}.$$

Hence

$$\Phi_j^+(t) \leq C_s M^3 2^{(3-2s)j}.$$

Now return to the definition of D_j^{res} :

$$D_j^{\text{res}}(t; t_0) = 2^{2j} \int_{t_0}^t e^{-c_0 \nu 2^{2j}(t-\sigma)} \Phi_j^+(\sigma) d\sigma.$$

Using the source bound just proved,

$$D_j^{\text{res}}(t; t_0) \leq 2^{2j} C_s M^3 2^{(3-2s)j} \int_{t_0}^t e^{-c_0 \nu 2^{2j}(t-\sigma)} d\sigma.$$

The remaining integral is controlled by the full heat tail:

$$\int_{t_0}^t e^{-c_0 \nu 2^{2j}(t-\sigma)} d\sigma \leq \int_0^\infty e^{-c_0 \nu 2^{2j}\rho} d\rho.$$

The last integral is explicit:

$$\int_0^\infty e^{-c_0 \nu 2^{2j}\rho} d\rho = \frac{1}{c_0 \nu 2^{2j}}.$$

Substituting this gives

$$D_j^{\text{res}}(t; t_0) \leq 2^{2j} C_s M^3 2^{(3-2s)j} \frac{1}{c_0 \nu 2^{2j}}.$$

The two factors 2^{2j} cancel:

$$D_j^{\text{res}}(t; t_0) \leq \frac{C_s}{c_0 \nu} M^3 2^{(3-2s)j}.$$

Set

$$C'_s = \frac{C_s}{c_0 \nu}.$$

This constant is independent of j , so

$$D_j^{\text{res}}(t; t_0) \leq C'_s M^3 2^{(3-2s)j}.$$

It remains to check the exponent. Since $s > 5/2$,

$$2s > 5.$$

Subtracting from 3 gives

$$3 - 2s < 3 - 5 = -2.$$

In particular,

$$3 - 2s < 0.$$

Set

$$a = 2s - 3.$$

Then $a > 0$ and $3 - 2s = -a$. Hence

$$2^{(3-2s)j} = 2^{-aj} = \frac{1}{2^{aj}}.$$

As $j \rightarrow \infty$, $2^{aj} \rightarrow \infty$, so

$$2^{(3-2s)j} \rightarrow 0.$$

The constants C'_s and M are fixed while $j \rightarrow \infty$. Thus

$$C'_s M^3 2^{(3-2s)j} \rightarrow 0.$$

The bound

$$0 \leq D_j^{\text{res}}(t; t_0) \leq C'_s M^3 2^{(3-2s)j}$$

then gives

$$D_j^{\text{res}}(t; t_0) \rightarrow 0.$$

This proves the proposition.

Contradiction with a selected terminal witness

Suppose an annular or funnel selection produces time windows and shell indices

$$j_m \rightarrow \infty.$$

The index m labels the selected window. Its end time is t_m , its length is τ_m , and its start time is $t_m - \tau_m$. The selected residual refill is

$$D_{j_m}^{\text{res}}(t_m; t_m - \tau_m).$$

The proposition gives

$$D_{j_m}^{\text{res}}(t_m; t_m - \tau_m) \leq C'_s M^3 2^{(3-2s)j_m}.$$

Since $j_m \rightarrow \infty$ and $3 - 2s < 0$, the right hand side tends to zero. For each fixed $\eta > 0$, there is an index m_0 such that, for every $m \geq m_0$,

$$C'_s M^3 2^{(3-2s)j_m} < \eta.$$

For those m ,

$$D_{j_m}^{\text{res}}(t_m; t_m - \tau_m) < \eta.$$

A fixed positive witness asserts the opposite lower bound along the same selected sequence:

$$D_{j_m}^{\text{res}}(t_m; t_m - \tau_m) \geq \eta.$$

For large m , the same number is smaller than η and at least η . The contradiction shows that a uniform H^s bound with $s > 5/2$ cannot coexist with a fixed positive residual refill witness at frequencies $j_m \rightarrow \infty$.